Mohamed Awadalla

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Experience

New York City Emergency Management Jun 2025 – Present
Software Development & Automation Intern *Brooklyn, NY*

- Built a JavaScript integration with the Zoom Events REST API to automate registration and attendance workflows for hybrid events serving 1,000+ registrants; added structured logging and failed-request tracking to support operational recovery
- Developed an OKR tracking application using Power Apps, SharePoint, and Power Automate, centralizing 63 active objectives and reporting workflows for the Office of the Chief Counsel
- Automated quarterly reminders, compliance follow-ups, and HTML leadership reports through three Power Automate workflows with retry and failed-item recovery
- Evaluated eight legal matter management platforms across technical capability, security, cost, and integration requirements; presented a final recommendation to the Chief Counsel

Education

Long Island University – Honors College May 2026
Bachelor of Science in Computer Science *Brooklyn, NY*
Honors: Dean Scholar, Dean's List

Projects

LegalDocuMan | *Python, Flask, PostgreSQL, Redis, RQ, React, RF-DETR, OCR, Docker* GitHub

- Built a full-stack contract-processing platform with a Flask REST API and React dashboard for OCR, document classification, metadata extraction, signature detection, and retention-policy assignment
- Designed a modular processing pipeline with interchangeable OCR, storage, and extraction backends; used Redis and RQ workers to move long-running OCR and inference jobs outside the API request cycle
- Fine-tuned an RF-DETR object-detection model on a self-curated dataset of 3,996 document images, achieving 94.9% mAP@50 for signature detection and using the detected signatures to determine document execution status

KnicksIQ | *Python, FastAPI, React, PostgreSQL, Redis, Qdrant, Docker, Render* GitHub

- Built and deployed a Knicks analytics application with a React frontend, FastAPI backend, and PostgreSQL database for exploring season and game data
- Designed a grounded-answer pipeline combining structured-table and lexical retrieval to return concise answers with supporting evidence and source-game references
- Implemented asynchronous NBA data ingestion with Redis and RQ and containerized the API, worker, database, vector store, and frontend using Docker Compose

Custom Autograd Engine & Character-Level Language Model | *Python, PyTorch* GitHub

- Implemented a reverse-mode automatic differentiation engine with tensor operations, topological backpropagation, and finite-difference gradient checks; validated computed gradients against PyTorch autograd
- Built and trained a character-level neural language model with embeddings and multilayer perceptron components on a dataset of 32,000 names

Technical Skills

Languages: Python, JavaScript, TypeScript, SQL, C++
Backend: Flask, FastAPI, Node.js, REST APIs, Redis, RQ
Data: PostgreSQL, Qdrant
Frontend: React, Vite
AI/ML: PyTorch, RF-DETR, OCR, Embeddings, Model Evaluation
Infrastructure: Docker, Docker Compose, Git, Linux, Render, Cloudflare